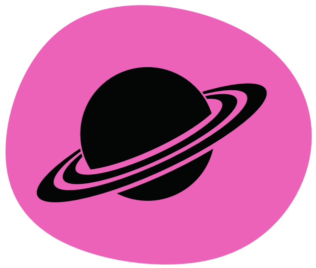
Fun Facts



The mummichog was the first fish sent to space!

This fish can survive in water that is 4x saltier than seawater!

Mummichogs do not have a stomach! Their esophagus connects directly to their intestine.

Interested in learning more?

Contact Us

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WHITEHEAD
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ECOLOGICAL GENOMICS IN THE ANTHROPOCENE

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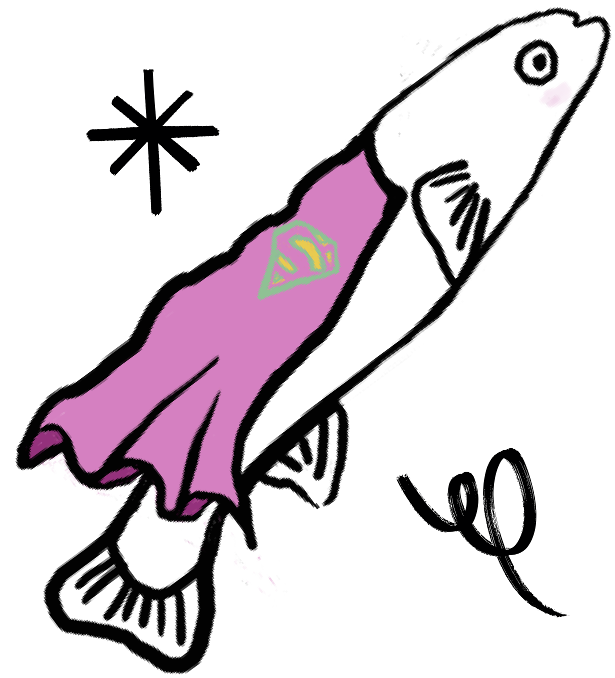


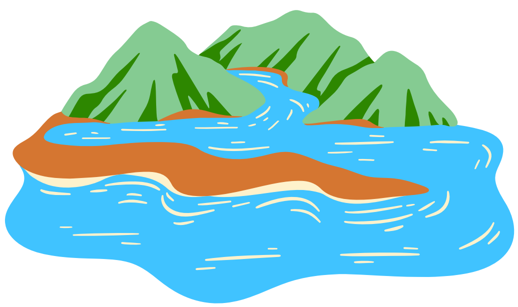
Mighty Mummichog

Superhero of the salt marsh

UC Davis Picnic Day 2025

What is a Mummichog?

Mummichogs, or Atlantic killifish, are about the size of your thumb. These killifish live in estuaries on the east coast. Estuaries are habitats where freshwater and saltwater meet and mix. Salinity in these environments change frequently with tides and rainfall, which is challenging for most organisms.



Though unassuming, killifish have a **superpower** that allows them to be one of the most abundant fish species in estuaries on the east coast:

~ **Osmo/Ionoregulatory Plasticity** ~
Flexibility to control the amount of water and salt entering and exiting so they can maintain homeostasis

Killifish do this by quickly and reversibly restructuring the cells in their gills to let them adjust to changes in salinity!

What do we hope to learn?

Key Terms:

Salinity = concentration of dissolved salts in water

Homeostasis = maintenance of stable internal body conditions important for organism survival

Genome = all the genetic material in an organism

Our world is increasingly characterized by extreme enrivonmental disturbances. By asking what parts of the genome responsible for this superpower, we can learn about the ability of organisms to adapt to change.

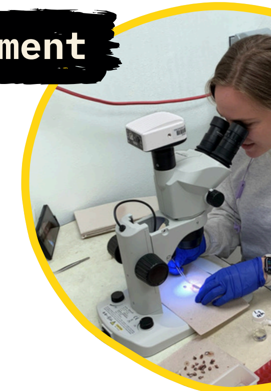
Exciting Questions to Answer

- Which regions of the genome are important for this superpower?
- For the fish that do not have the superpower, do they lose this ability in unique ways?
- Are there particular genetic changes that might help us learn about other types of similar superpowers in other animals?

How are we doing it?

We answer our questions by comparing the DNA of killifish with the superpower to those without it.

Salinity Experiment



I ran an experiment where I exposed killifish with the superpower and those without it to different salinities. Then I dissected the gills from these fish so I could run molecular tests to understand which genes are being used to respond to changes in salinity and how these differ between species.

Gill Microscopy

The surface of killifish gills look like fingerprints with holes in seawater.

In freshwater, these holes look like fingers instead.

We compare differences in the gill surface to learn in what situations mummichogs are using their superpower.

Compare Genomes

To analyze differences in genome sequences between species, we use a lot of computer power to wrangle millions of pieces of information.